Trade with Variable Markups

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Trade, Competition, and Markups

- Krugman (1979): with a demand elasticity that rises with price, trade lowers markups and raises firm scale (a "pro-competitive" effect)
- Krugman (1980), Melitz (2003): CES + monopolistic competition $\Rightarrow$ constant markup $\frac{\sigma}{\sigma-1}$; the dominant paradigm in trade
- Under CES, trade affects welfare through variety, selection and reallocation, but never through markups

Two routes to restore variable markups

1. Monopolistic competition with **non-CES demand**: Mrázová and Neary (2017); Melitz and Ottaviano (2008); Feenstra and Weinstein (2017); Arkolakis, Costinot, Donaldson and Rodríguez-Clare (2019)
2. **Oligopoly** with a finite number of firms per market: Atkeson and Burstein (2008); Edmond, Midrigan and Xu (2015)

Central question: are the gains from trade *larger* once markups respond to competition?

Outline

1. **Beyond CES: Demand Structure**
2. Linear Demand: Melitz and Ottaviano (2008)
3. Translog: Feenstra and Weinstein (2017)
4. The Elusive Pro-Competitive Effects: ACDR (2019)
5. Oligopoly: Atkeson and Burstein (2008)
6. Edmond, Midrigan and Xu (2015)

Mrázová and Neary (2017): Elasticity and Convexity

- Inverse demand $p(x)$, constant marginal cost c: $\max_x \; p(x)x - cx$

$$\text{FOC: } p + xp'(x) = c, \qquad \text{SOC: } 2p'(x) + xp''(x) < 0$$

- Two local properties of demand

$$\text{elasticity } \varepsilon(x) \equiv -\frac{p(x)}{xp'(x)}, \qquad \text{convexity } \rho(x) \equiv -\frac{xp''(x)}{p'(x)}$$

- FOC $\Leftrightarrow$ markup $\dfrac{p}{c} = \dfrac{\varepsilon}{\varepsilon - 1}$; SOC $\Leftrightarrow \rho < 2$ (divide by $p' < 0$)
- As x varies, each demand function traces a curve in (ε, ρ) space: its **demand manifold**
  - The manifold is a sufficient statistic for many comparative statics (pass-through, selection, response to market size)

Demand Manifolds of Common Demand Systems

Linear (Melitz–Ottaviano), translog (Feenstra–Weinstein), CES and others. Source: Mrázová and Neary (2017).

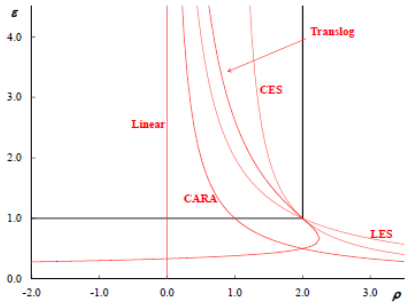


Figure 2: Demand Manifolds for Some Common Demand Functions

When Do Markups Vary Across Firms?

- Markups vary with firm size only if ε varies with x. Differentiating $\varepsilon = -p/(xp')$:

$$\varepsilon'(x) = -\frac{1}{x} - \frac{\varepsilon}{x} + \frac{p\,p''}{x\,(p')^2} = \frac{\varepsilon}{x}\left(\rho - \frac{\varepsilon+1}{\varepsilon}\right)$$

  using $\frac{p\,p''}{x(p')^2} = \frac{p}{xp'} \cdot \frac{p''}{p'} = (-\varepsilon)\left(-\frac{\rho}{x}\right)$

- CES, $p = x^{-1/\sigma}$: $\varepsilon = \sigma$ and $\rho = \frac{\sigma+1}{\sigma} = \frac{\varepsilon+1}{\varepsilon}$, so $\varepsilon'(x) \equiv 0$: no markup variation
- **Sub-convex** demand, $\rho < \frac{\varepsilon+1}{\varepsilon}$ (less convex than the CES with the same elasticity): $\varepsilon'(x) < 0$
  - Larger (lower-cost) firms face less elastic demand and charge *higher* markups
  - Equivalently, demand becomes more elastic as price rises: "Marshall's Second Law of Demand" (Melitz, 2018)
  - Linear, translog, CARA and Stone–Geary (positive subsistence) demands are sub-convex
- Super-convex demand ($\rho > \frac{\varepsilon+1}{\varepsilon}$) delivers the opposite and is generally viewed as empirically implausible

Sub-convex and Super-convex Regions

The CES locus $\rho = \frac{\varepsilon+1}{\varepsilon}$ divides the (ε, ρ) plane. Source: Mrázová and Neary (2017).

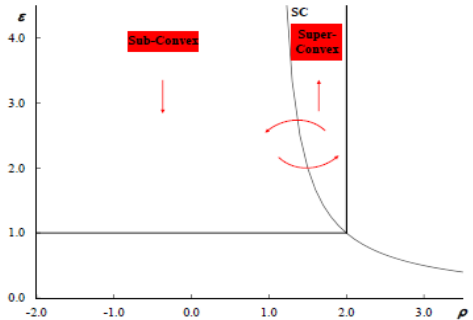


Figure 4: The Sub- and Super-Convex Regions

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Melitz and Ottaviano (2008): Quadratic Preferences

- Quasi-linear quadratic utility over a numeraire q_0^c and varieties $i \in \Omega$

$$U = q_0^c + \alpha \int_{i\in\Omega} q_i^c \, di - \tfrac{1}{2}\gamma \int_{i\in\Omega} (q_i^c)^2 di - \tfrac{1}{2}\eta \left(\int_{i\in\Omega} q_i^c \, di \right)^2$$

  γ: degree of product differentiation; α, η: substitution between varieties and the numeraire

- Inverse demand: $p_i = \alpha - \gamma q_i^c - \eta Q^c$, $Q^c = \int_{i\in\Omega} q_i^c \, di$
- Let $\Omega^* \subseteq \Omega$ be the varieties consumed, N their measure and $\bar{p} = \frac{1}{N}\int_{\Omega^*} p_i \, di$. Integrating the inverse demand over Ω^*: $Q^c = \frac{N(\alpha - \bar{p})}{\gamma + \eta N}$. Hence market demand ($L$ consumers)

$$q_i \equiv L q_i^c = \frac{L}{\gamma}\left(p_{\max} - p_i\right), \qquad p_{\max} \equiv \frac{\gamma\alpha + \eta N \bar{p}}{\gamma + \eta N}$$

- Linear demand has a **choke price** $p_{\max}$, which falls with more ($N \uparrow$) or cheaper ($\bar{p} \downarrow$) competitors

Pricing, Cutoff and Free Entry (Closed Economy)

- Demand elasticity $\varepsilon_i = \frac{L}{\gamma}\frac{p_i}{q_i} = \frac{p_i}{p_{\max}-p_i}$, rising in p_i (sub-convex). Monopoly pricing, $\frac{p}{c} = \frac{\varepsilon}{\varepsilon-1} = \frac{p}{2p-p_{\max}}$, gives

$$p(c) = \tfrac{1}{2}(c_D + c), \qquad q(c) = \tfrac{L}{2\gamma}(c_D - c), \qquad \pi(c) = \tfrac{L}{4\gamma}(c_D - c)^2$$

  where $c_D = p_{\max}$ is the cost cutoff. Markup $p(c) - c = \frac{1}{2}(c_D - c)$: lower-cost firms charge higher markups

- Free entry with sunk cost f_E and cost draws $c \sim G(c) = (c/c_M)^k$ on $[0, c_M]$:

$$\int_0^{c_D} \pi(c)\, dG(c) = f_E \;\Rightarrow\; c_D = \left[\frac{2(k+1)(k+2)\gamma\, c_M^k f_E}{L}\right]^{\frac{1}{k+2}}$$

- Number of sellers from the choke price, using $\bar{p} = \frac{1}{2}(c_D + \bar{c})$ and $\bar{c} = \frac{k}{k+1}c_D$:

$$c_D = \frac{\gamma\alpha + \eta N\bar{p}}{\gamma + \eta N} \;\Rightarrow\; N = \frac{2\gamma}{\eta}\frac{\alpha - c_D}{c_D - \bar{c}} = \frac{2(k+1)\gamma}{\eta}\frac{\alpha - c_D}{c_D}$$

- Entrants $N_E = N/G(c_D)$ are endogenous; only N matters for prices and welfare

Market Size, Trade and Welfare

- Substituting demands into U (with I^c consumer income and σ_p^2 the price variance across varieties)

$$U = I^c + \tfrac{1}{2}\left(\eta + \tfrac{\gamma}{N}\right)^{-1}(\alpha - \bar{p})^2 + \tfrac{1}{2}\tfrac{N}{\gamma}\sigma_p^2 \overset{\text{Pareto}}{=} 1 + \tfrac{1}{2\eta}(\alpha - c_D)\left(\alpha - \tfrac{k+1}{k+2}c_D\right)$$

  so welfare is decreasing in c_D

- Larger market $L \uparrow$ (as in Krugman 1979): $c_D \downarrow \Rightarrow$ tougher selection, higher average productivity, lower markups, more varieties N, higher welfare
- Open economy: exporters to h face cutoff $c_X = c_D^h/\tau$. Two symmetric countries, freeness $\phi \equiv \tau^{-k}$:

$$c_D = \left[\frac{2(k+1)(k+2)\gamma\, c_M^k f_E}{L\,(1+\phi)}\right]^{\frac{1}{k+2}}$$

  trade liberalization ($\phi \uparrow$) acts like an increase in market size (asymmetric case: MO eq. (23))
- Does trade liberalization deliver *more* welfare gains than in a CES model with the same trade data? $\Rightarrow$ ACDR below

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Feenstra and Weinstein (2017): Symmetric Translog

- Symmetric translog expenditure function over N *potential* varieties ($\sum_i \alpha_i = 1$)

$$\ln e_t = \alpha_0 + \sum_i \alpha_i \ln p_{it} - \frac{\gamma}{2} \sum_i (\ln p_{it})^2 + \frac{\gamma}{2N} \sum_i \sum_j \ln p_{it} \ln p_{jt}$$

- Shephard's lemma gives expenditure shares

$$s_{it} = \frac{\partial \ln e_t}{\partial \ln p_{it}} = \alpha_i - \gamma \left(\ln p_{it} - \overline{\ln p}(t) \right), \qquad \overline{\ln p}(t) \equiv \frac{1}{N} \sum_{j=1}^{N} \ln p_{jt}$$

- Unavailable goods ($i \notin I_t$, N_t goods available) enter at their reservation price, $s_{it} = 0$: $\ln \tilde{p}_{it} = \frac{\alpha_i}{\gamma} + \overline{\ln p}(t)$. Solving out:

$$\overline{\ln p}(t) = \underbrace{\frac{1}{N_t} \sum_{i \in I_t} \ln p_{it}}_{\overline{\ln p}_t\text{: average over available goods}} + \frac{1}{\gamma} \underbrace{\frac{1}{N_t} \left(1 - \sum_{i \in I_t} \alpha_i \right)}_{\bar{\alpha}_t\text{: falls as goods enter}}$$

Cost of Living, Variety and Markups

- Invert shares, $\ln p_{it} = \overline{\ln p}(t) + (\alpha_i - s_{it})/\gamma$, and substitute into e_t:

$$\ln e_t = \alpha_0 + \tfrac{1}{2\gamma}\textstyle\sum_{i=1}^{N} \alpha_i^2 + \overline{\ln p}(t) - \tfrac{1}{2\gamma}\textstyle\sum_{i=1}^{N} s_{it}^2$$

- Using the expression for $\overline{\ln p}(t)$ and differencing over time:

$$\Delta \ln e_t = \underbrace{\Delta\overline{\ln p}_t}_{\text{prices (incl. markups)}} + \underbrace{\frac{\Delta\bar{\alpha}_t}{\gamma}}_{\text{variety}} \underbrace{- \frac{1}{2\gamma}\Delta HHI_t}_{\text{“crowding”}}$$

  - Entry of new goods lowers $\bar{\alpha}_t$ (gain); it also lowers HHI_t, which *raises* e_t: in a more crowded product space each variety is valued less

- Pro-competitive effect through prices: own-price elasticity $1 + \frac{\gamma(N_t - 1)}{N_t s_{it}}$ gives $\ln p_{it} = \ln c_{it} + \ln\left[1 + \frac{s_{it} N_t}{\gamma(N_t - 1)}\right]$: markups rise with market share and fall with the number of competitors N_t
- Without firm-level data, FW use Herfindahl indexes by source country. U.S. 1992–2005: welfare gain $\approx 0.86\%$, about half from new varieties and half from lower markups

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Arkolakis, Costinot, Donaldson and Rodríguez-Clare (2019)

- A general class of demand systems with variable markups

$$q_\omega(\mathbf{p}, w) = Q(\mathbf{p}, w)\, D\left(\frac{p_\omega}{P(\mathbf{p}, w)}\right)$$

  with aggregate shifters Q, P solving, for $\beta \in \{0, 1\}$ and H increasing and concave,

$$\int_\omega \left[H\left(\tfrac{p_\omega}{P}\right)\right]^\beta \left[p_\omega Q D\left(\tfrac{p_\omega}{P}\right)\right]^{1-\beta} d\omega = w^{1-\beta}, \qquad Q^{1-\beta}\left[\int_\omega p_\omega Q D\left(\tfrac{p_\omega}{P}\right) d\omega\right]^\beta = w^\beta$$

  - $\beta = 0$: additively separable utility (Krugman 1979), non-homothetic; $Q \equiv 1$
  - $\beta = 1$: symmetric translog/QMOR and Kimball (1995), homothetic

- A1 (**choke price**): $D(x) = 0$ for all $x \geq 1$, so P is the choke price
- A2 (**Pareto**): $G_i(z) = 1 - (b_i/z)^\theta$, common θ
- Monopolistic competition, free entry (cost $w_i F_i$), iceberg τ_{ij}, *no* fixed market-access costs: selection into markets only through the choke price

Firm-Level Markups and Sales

- Firm with productivity z from i selling to j: marginal cost $c = w_i \tau_{ij}/z$. With $\varepsilon_D(x) \equiv -\frac{d \ln D(x)}{d \ln x}$, the markup $m \equiv p/c$ solves

$$m = \frac{\varepsilon_D(m/v)}{\varepsilon_D(m/v) - 1}, \qquad v \equiv \frac{P_j}{c} \text{ (efficiency relative to the choke price)}$$

  so $m = \mu(v)$, with $\mu(1) = 1$; $\mu'(v) > 0$ if demand is log-concave in log price (sub-convex)

- Cutoff: j buys from firms with $c \leq P_j$, i.e. $z \geq z^*_{ij} \equiv w_i \tau_{ij}/P_j$; hence $v = z/z^*_{ij}$
- Firm sales in j (population L_j)

$$x(c, v, Q_j, L_j) = L_j Q_j \, c \, \mu(v) \, D\left(\frac{\mu(v)}{v}\right)$$

- Profits are $\frac{\mu(v)-1}{\mu(v)} \, x$: same as CES, except that the markup varies across firms

First Key Result: Gravity Holds

- Aggregate sales from i to j (normalize $b_i = 1$, so $dG_i(z) = \theta z^{-\theta-1} dz$):

$$X_{ij} = N_i \int_{z_{ij}^*}^{\infty} L_j Q_j \frac{w_i \tau_{ij}}{z} \mu\left(\frac{z}{z_{ij}^*}\right) D\left(\frac{\mu(z/z_{ij}^*)}{z/z_{ij}^*}\right) \theta z^{-\theta-1} dz$$

- Change variables $v = z/z_{ij}^*$ and use $z_{ij}^* = w_i \tau_{ij}/P_j$:

$$X_{ij} = \chi\, N_i\, (w_i \tau_{ij})^{-\theta}\, L_j Q_j\, P_j^{1+\theta}, \qquad \chi \equiv \theta \int_1^{\infty} \mu(v)\, D\left(\frac{\mu(v)}{v}\right) v^{-\theta-2} dv$$

- Similarly, gross profits $\Pi_{ij} = (\pi/\chi)\, X_{ij}$ for a constant π: a constant share of sales
- Free entry ($\sum_j \Pi_{ij} = w_i F_i N_i$) + labor market clearing ($\sum_j X_{ij} = w_i L_i$) $\Rightarrow N_i = \frac{\pi}{\chi} \frac{L_i}{F_i}$, invariant to trade costs
- Hence $\lambda_{ij} = \frac{N_i (w_i \tau_{ij})^{-\theta}}{\sum_k N_k (w_k \tau_{kj})^{-\theta}}$: the same gravity equation and the same counterfactual wages and trade flows as in ACR (2012), with trade elasticity θ

Welfare: Shephard's Lemma

- Numeraire $w_j = 1$; free entry $\Rightarrow$ income per capita is w_j, so $d \ln W_j = -d \ln e_j$
- Shephard's lemma, $\partial e_j / \partial p_{\omega j} = q_{\omega j}$:

$$d \ln e_j = \sum_i \int_{\omega \in \Omega_{ij}} \lambda_{\omega j} \, d \ln p_{\omega j} \, d\omega = \sum_i \int_{z_{ij}^*}^{\infty} \lambda_{ij}(z) \left[d \ln c_{ij} + d \ln m_{ij}(z) \right] dG_i(z)$$

with $\lambda_{\omega j}$ the expenditure share on ω, $c_{ij} \equiv w_i \tau_{ij}$, $m_{ij}(z) = \mu(z / z_{ij}^*)$

  - Goods entering or exiting at the cutoff are priced at the choke price: no first-order welfare effect (no gains from new varieties here)

- Since z is fixed, $d \ln m_{ij}(z) = \frac{d \ln \mu(v)}{d \ln v} \, d \ln v = - \frac{d \ln \mu(v)}{d \ln v} \, d \ln z_{ij}^*$. Therefore

$$d \ln e_j = \sum_i \lambda_{ij} \left[d \ln c_{ij} - \rho_{ij} \, d \ln z_{ij}^* \right], \qquad \rho_{ij} \equiv \int_{z_{ij}^*}^{\infty} \frac{\lambda_{ij}(z)}{\lambda_{ij}} \frac{d \ln \mu(v)}{d \ln v} \Big|_{v = z / z_{ij}^*} dG_i(z)$$

where $\lambda_{ij} = X_{ij} / e_j$; $\rho_{ij} \geq 0$ is a sales-weighted average markup elasticity

Second Key Result: Direct vs. Indirect Markup Effects

- Under Pareto the weights depend only on v:

$$\frac{\lambda_{ij}(z)}{\lambda_{ij}} = \frac{(z_{ij}^*)^{1+\theta}}{\chi}\,\mu\Big(\tfrac{z}{z_{ij}^*}\Big)\,D\Big(\tfrac{\mu(z/z_{ij}^*)}{z/z_{ij}^*}\Big)\,\frac{1}{z} \;\Rightarrow\; \rho_{ij} = \rho \equiv \frac{\int_1^\infty \frac{d\ln\mu}{d\ln v}\,\mu(v)D\Big(\frac{\mu(v)}{v}\Big)\,v^{-\theta-2}dv}{\int_1^\infty \mu(v)D\Big(\frac{\mu(v)}{v}\Big)\,v^{-\theta-2}dv}$$

- Substituting $d\ln z_{ij}^* = d\ln c_{ij} - d\ln P_j$ and $\sum_i \lambda_{ij} = 1$:

$$d\ln e_j = \underbrace{\sum_i \lambda_{ij} d\ln c_{ij}}_{\text{marginal costs}} \underbrace{- \rho \sum_i \lambda_{ij} d\ln c_{ij}}_{\text{direct markup effect}} + \underbrace{\rho\, d\ln P_j}_{\text{indirect markup effect}}$$

- For a "good" shock ($\sum_i \lambda_{ij} d\ln c_{ij} < 0$) with $\rho > 0$
  - **Direct**: foreign exporters whose costs fall raise their markups (incomplete pass-through) $\Rightarrow$ smaller gains
  - **Indirect** (pro-competitive): tougher competition lowers P_j and all markups in j $\Rightarrow$ larger gains
- Which effect dominates depends on how P_j responds to c_{ij}

Solving for the Choke Price

- With Pareto and $w_j = 1$, the two equations defining Q_j, P_j become

$$Q_j^{1-\beta} P_j^{1+\theta-\beta} \kappa_1 \sum_i N_i c_{ij}^{-\theta} = 1, \qquad Q_j \, P_j^{\beta(1+\theta)} \left(\kappa_2 \sum_i N_i c_{ij}^{-\theta} \right)^{\beta} = 1$$

  for constants κ_1, κ_2 (same change of variables, $v = z/z_{ij}^*$). $\beta = 0$: $Q_j = 1$; $\beta = 1$: first equation alone pins down P_j. Either way

$$P_j = \left(\kappa \sum_i N_i c_{ij}^{-\theta} \right)^{-\frac{1}{1+\theta-\beta}}$$

- Total differentiation, with N_i fixed and $\frac{N_i c_{ij}^{-\theta}}{\sum_k N_k c_{kj}^{-\theta}} = \lambda_{ij}$:

$$d \ln P_j = \frac{\theta}{1+\theta-\beta} \sum_i \lambda_{ij} \, d \ln c_{ij}$$

- $\beta = 1$ (homothetic): $d \ln P_j = \sum_i \lambda_{ij} d \ln c_{ij}$, so direct and indirect effects exactly offset
- $\beta = 0$ (additively separable): $d \ln P_j = \frac{\theta}{1+\theta} \sum_i \lambda_{ij} d \ln c_{ij}$, so the direct (anti-competitive) effect dominates

ACDR: The New Welfare Formula

- Combining,

$$d \ln e_j = (1-\eta) \sum_i \lambda_{ij} \, d \ln c_{ij}, \qquad \eta \equiv \rho \, \frac{1-\beta}{1+\theta-\beta}$$

- From gravity, $d \ln \lambda_{ij} - d \ln \lambda_{jj} = -\theta \, d \ln c_{ij}$ (as $c_{jj} = 1$ is fixed); multiply by λ_{ij} and sum over i: $\sum_i \lambda_{ij} d \ln c_{ij} = d \ln \lambda_{jj} / \theta$
- Proposition 1:

$$d \ln W_j = -(1-\eta) \, \frac{d \ln \lambda_{jj}}{\theta}$$

  - $\beta = 1$ (translog, Kimball): $\eta = 0$, exactly the ACR gains
  - $\beta = 0$ with sub-convex demand ($\rho > 0$): $\eta > 0$, gains *smaller* than ACR
- Pareto also makes the univariate markup distribution invariant to trade costs: incumbent exporters raise markups while new low-markup exporters enter; what matters for welfare is the joint distribution of markups and employment
- Estimation on U.S. micro trade data gives a small positive η: pro-competitive effects are “elusive”

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Oligopoly, Trade and Markups

1. Atkeson and Burstein (2008): variable markups from oligopoly
   - Nested CES, finite number of firms per sector, Cournot competition
   - Originally a two-country model of pricing-to-market; here we use a closed-economy version to study markup dispersion and misallocation

2. Edmond, Midrigan and Xu (2015): the same structure in an open economy
   - Quantifies the pro-competitive gains from trade
   - Key role for head-to-head competition between domestic and foreign firms

Key features of the AB structure

- Demand elasticity faced by a firm *decreases* with its market share
- Reallocating market shares changes both the level and the dispersion of markups, and therefore aggregate TFP

Nested CES and Demand

- Continuum of sectors, n firms per sector

$$Y = \left(\int_0^1 y(s)^{\frac{\theta-1}{\theta}} \, ds \right)^{\frac{\theta}{\theta-1}}, \qquad y(s) = \left(\sum_{i=1}^{n} y_i(s)^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}}, \qquad \gamma > \theta > 1$$

  (firms within a sector are closer substitutes than goods across sectors)

- Competitive final-good producers maximize $PY - \int_0^1 \sum_i p_i(s) y_i(s) \, ds$:

$$y_i(s) = \left(\frac{p_i(s)}{p(s)} \right)^{-\gamma} \left(\frac{p(s)}{P} \right)^{-\theta} Y$$

$$p(s) = \left(\sum_{i=1}^{n} p_i(s)^{1-\gamma} \right)^{\frac{1}{1-\gamma}}, \qquad P = \left(\int_0^1 p(s)^{1-\theta} \, ds \right)^{\frac{1}{1-\theta}}$$

- Technology $y_i(s) = a_i(s) \, l_i(s)^{\alpha} k_i(s)^{1-\alpha}$, marginal cost $c(w,r)/a_i(s)$; productivity IID Pareto, $\Pr[a_i(s) \geq a] = a^{-\xi}$ (thick tails if ξ low)
- No *ex ante* sectoral heterogeneity, but *ex post* heterogeneity because each sector has only n draws

Cournot Pricing

- Price is a markup over marginal cost

$$p_i(s) = \mu_i(s)\,\frac{c(w,r)}{a_i(s)}, \qquad \mu_i(s) = \frac{\varepsilon_i(s)}{\varepsilon_i(s)-1}$$

- Under Cournot competition, the perceived demand elasticity is a share-weighted harmonic mean of θ and γ

$$\varepsilon_i(s) = \left(\omega_i(s)\frac{1}{\theta} + \big(1-\omega_i(s)\big)\frac{1}{\gamma}\right)^{-1}, \qquad \omega_i(s) \equiv \frac{p_i(s)y_i(s)}{\sum_k p_k(s)y_k(s)} = \left(\frac{p_i(s)}{p(s)}\right)^{1-\gamma}$$

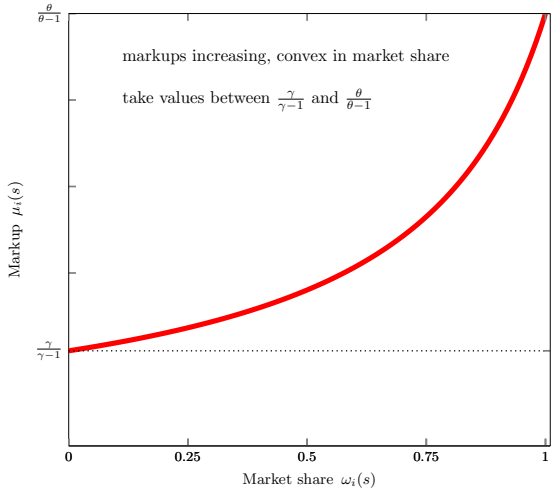
- Equivalently, the inverse markup is linear in market share

$$\frac{1}{\mu_i(s)} = \frac{\gamma-1}{\gamma} - \left(\frac{1}{\theta}-\frac{1}{\gamma}\right)\omega_i(s)$$

  markups rise from $\frac{\gamma}{\gamma-1}$ (as $\omega \to 0$) to $\frac{\theta}{\theta-1}$ (as $\omega \to 1$)

- Equilibrium: fixed point in prices, $\mathbf{p} = h\big(g(f(\mathbf{p})), \mathbf{a}\big)$ with $\boldsymbol{\omega} = f(\mathbf{p})$, $\boldsymbol{\varepsilon} = g(\boldsymbol{\omega})$; then recover shares and markups

Markups $\mu_i(s)$ and Market Shares $\omega_i(s)$



Aggregate TFP and Misallocation

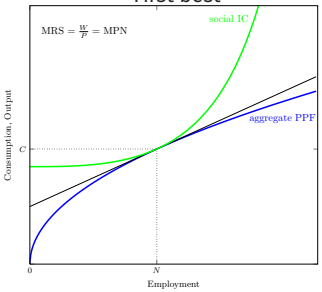
- Let $\mu(s) \equiv \frac{p(s)y(s)}{\text{cost}(s)}$ and $\mu \equiv \frac{PY}{\text{total cost}}$ be cost-weighted sectoral and aggregate markups
- Aggregate TFP (output per unit of the composite input)

$$A = \left(\int_0^1 \left(\frac{\mu(s)}{\mu} \right)^{-\theta} a(s)^{\theta-1} \, ds \right)^{\frac{1}{\theta-1}}, \qquad a(s) = \left(\sum_{i=1}^{n} \left(\frac{\mu_i(s)}{\mu(s)} \right)^{-\gamma} a_i(s)^{\gamma-1} \right)^{\frac{1}{\gamma-1}}$$

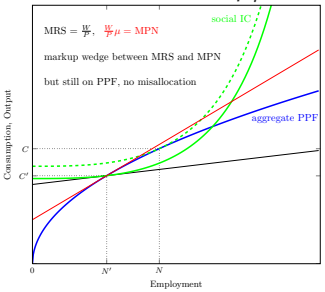
- First best (common markups): $a(s) = \left(\sum_i a_i(s)^{\gamma-1} \right)^{\frac{1}{\gamma-1}}$, $A = \left(\int_0^1 a(s)^{\theta-1} ds \right)^{\frac{1}{\theta-1}}$
  - With a common markup, relative prices equal relative marginal costs: $\frac{p_i(s)}{p_k(s)} = \frac{a_k(s)}{a_i(s)}$, so no misallocation
  - The *level* of the markup distorts the labor–leisure/capital margins, but not TFP
- Markup *dispersion* lowers A below first best: high-markup (large) firms produce too little

Markup Levels vs. Markup Dispersion

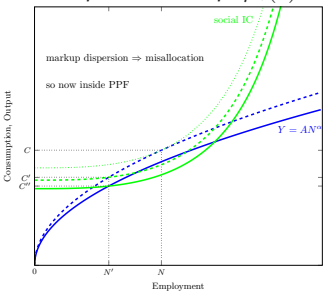
First best



Common markup μ



Dispersed markups $\mu_i(s)$



Homogeneous vs. Heterogeneous Firms

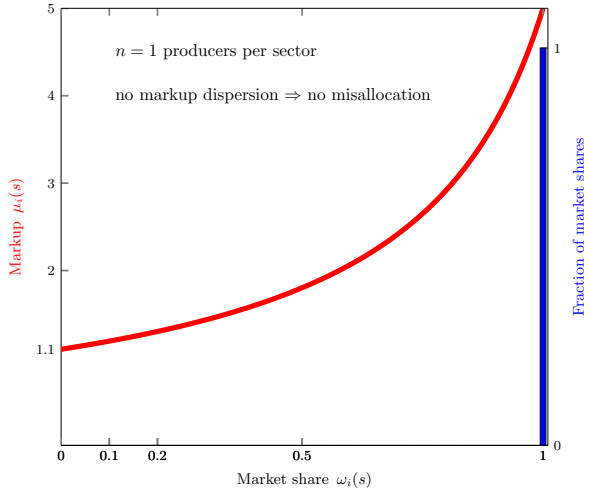
- **Homogeneous firms**: n identical producers, $\omega_i(s) = 1/n$

$$\mu = \frac{n}{\left(\frac{\gamma-1}{\gamma}\right)n - \left(\frac{1}{\theta} - \frac{1}{\gamma}\right)}$$

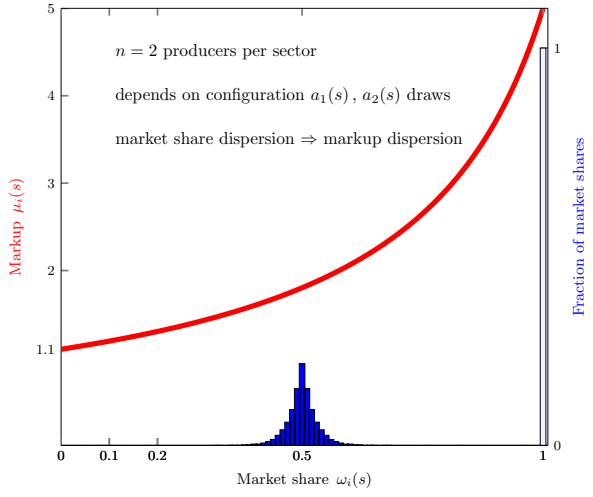
declining from $\frac{\theta}{\theta-1}$ at $n = 1$ to $\frac{\gamma}{\gamma-1}$ as $n \to \infty$; markup level distorts allocations, but TFP is at first best

- **Heterogeneous firms**
  - $n = 1$: monopoly markup $\frac{\theta}{\theta-1}$ (large), but no misallocation
  - $n = 2$: aggregate markup falls, but now markups are dispersed
  - Extent of dispersion depends on productivity dispersion
  - The fall in the aggregate markup also depends on dispersion (Jensen's inequality, as $\omega \mapsto \mu$ is convex)

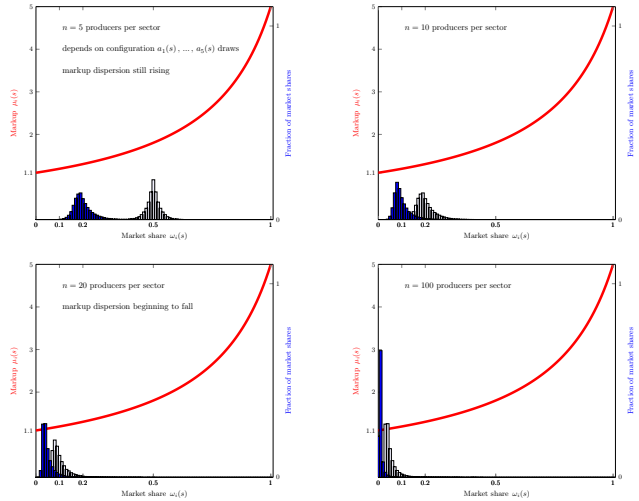
Monopoly: High Markup but No Misallocation



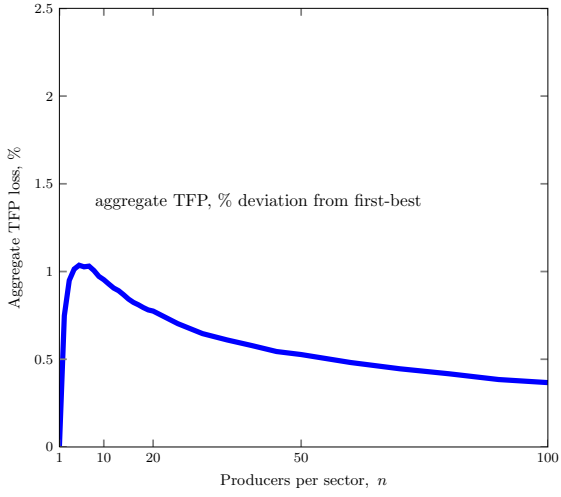
Duopoly: Lower Markups, but Now Misallocation



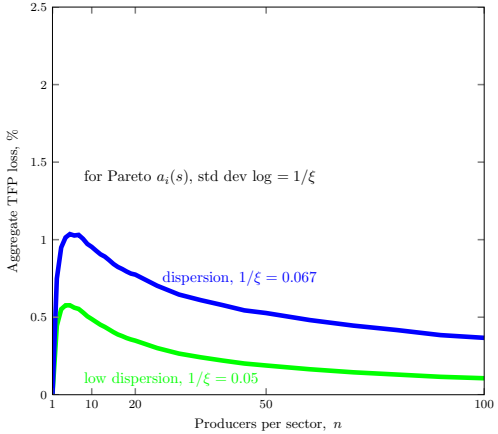
More Competitors: $n = 5, 10, 20, 100$

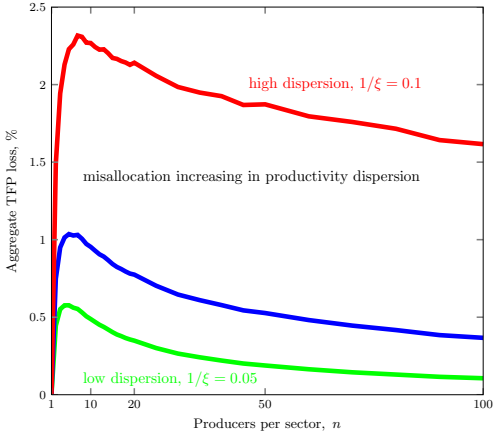


Misallocation and Competition



Misallocation and Productivity Dispersion





Summary: Closed-Economy Atkeson–Burstein

- Misallocation
  - Increasing in n for low n, then decreasing
  - Higher when productivity dispersion is higher (more chance of a dominant producer)
- Aggregate markup
  - Decreasing in n
- Implication for trade: foreign entry is like raising n, but the effect on misallocation depends on *who* the new competitors are and whom they compete with

Outline

1. Beyond CES: Demand Structure
2. Linear Demand: Melitz and Ottaviano (2008)
3. Translog: Feenstra and Weinstein (2017)
4. The Elusive Pro-Competitive Effects: ACDR (2019)
5. Oligopoly: Atkeson and Burstein (2008)
6. **Edmond, Midrigan and Xu (2015)**

Edmond, Midrigan and Xu (2015): Question

- Can trade significantly reduce product-market distortions?
- Trade affects welfare via two channels
  1. "Pro-competitive" effects: less markup dispersion $\Rightarrow$ less misallocation
  2. "Standard" effects: comparative advantage, love of variety, selection
- Pro-competitive effects are an old idea, with suggestive evidence from liberalization episodes, but models disagree
  - Bernard, Eaton, Jensen and Kortum (2003): Bertrand + Fréchet, markup distribution invariant to trade costs, *no* pro-competitive effects
  - ACDR (2019): *elusive* pro-competitive effects
- What matters is the response of the **joint distribution of markups and market shares**
- Approach: AB two-country model, disciplined with Taiwanese manufacturing data, adding sector heterogeneity, fixed costs to operate and to export

Main Findings

- Two key determinants of pro-competitive gains
  1. Initial **markup dispersion** determines the potential gains from reduced misallocation
  2. **Head-to-head competition** determines the extent to which those gains are realized

- Misallocation is large when there is
  - Extensive *concentration*, both within and across sectors
  - A low *across-sector* elasticity of substitution θ (dominant producers can raise markups)

  Taiwanese data: high concentration and low $\theta \Rightarrow$ substantial misallocation

- Head-to-head competition is governed by the *cross-country correlation* of productivity draws
  - High correlation $\Rightarrow$ weak comparative advantage $\Rightarrow$ high trade elasticity
  - Standard gravity estimates of the trade elasticity require a high correlation

Two-Country Model

- $n(s)$ potential producers per sector in each country

$$y(s) = \left(\sum_{i=1}^{n(s)} y_i^H(s)^{\frac{\gamma-1}{\gamma}} + \sum_{i=1}^{n(s)} y_i^F(s)^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}}, \qquad \gamma > \theta$$

- Home final-good producers pay iceberg trade cost $\tau \geq 1$ on foreign goods

$$y_i^H(s) = \left(\frac{p_i^H(s)}{p(s)} \right)^{-\gamma} \left(\frac{p(s)}{P} \right)^{-\theta} Y, \qquad y_i^F(s) = \left(\frac{\tau p_i^F(s)}{p(s)} \right)^{-\gamma} \left(\frac{p(s)}{P} \right)^{-\theta} Y$$

- Market share of a home firm in the home market (denominator includes imports)

$$\omega_i^H(s) = \frac{p_i^H(s) y_i^H(s)}{\sum_k p_k^H(s) y_k^H(s) + \tau \sum_k p_k^F(s) y_k^F(s)} = \left(\frac{p_i^H(s)}{p(s)} \right)^{1-\gamma}$$

- Same Cournot markup formula, market by market; operate domestically if profits cover f_d, export if export profits cover f_x
- Same marginal cost in both markets, but firms usually have smaller shares abroad $\Rightarrow$ more elastic demand and lower markups on exports

Aggregate Productivity and the Trade Elasticity

- Aggregate productivity has the AB form, with imports entering sector productivity

$$a(s) = \left(\sum_{i=1}^{n(s)} \left(\frac{\mu_i^H(s)}{\mu(s)} \right)^{-\gamma} a_i(s)^{\gamma-1} + \tau^{1-\gamma} \sum_{i=1}^{n(s)} \left(\frac{\mu_i^F(s)}{\mu(s)} \right)^{-\gamma} a_i^*(s)^{\gamma-1} \right)^{\frac{1}{\gamma-1}}$$

  Constant markups $\mu_i(s) = \mu$ give the usual CES benchmark

- Trade elasticity: response of relative spending to a q-fold increase in foreign prices

$$\sigma_{\text{rel. prices}} \equiv -\frac{d \log \frac{1-\lambda}{\lambda}}{d \log q} = (\gamma - 1) - (\gamma - \theta) \frac{\text{Var}[\lambda(s)]}{\lambda(1-\lambda)}$$

  $\lambda(s)$: domestic share in sector s; $\lambda = \int_0^1 \omega(s)\lambda(s)ds$ and Var use sector expenditure weights $\omega(s)$
  - Lies in $[\theta - 1, \gamma - 1]$; decreasing in import-share dispersion

- What we observe is $\sigma_{\text{trade costs}} \equiv -\frac{d \log \frac{1-\lambda}{\lambda}}{d \log \tau}$
  - Constant markups: $d \log q = d \log \tau$, the two coincide
  - Variable markups: incomplete pass-through, $d \log q < d \log \tau$, which dampens the response of trade flows to trade costs

Quantification: Strategy and Productivity Process

Three key ingredients and targets

1. Within-country productivity distribution $\rightarrow$ concentration facts
2. Gap between θ and γ: set $\gamma = 10$; choose θ to match the markup–share relation
3. Correlation of $a_i(s), a_i^*(s) \rightarrow$ trade elasticity

Productivity $a_i(s) = z(s)\, x_i(s)$

- Sector $z(s)$: IID Pareto(ξ_z)
- Idiosyncratic $x_i(s)$: IID Pareto(ξ_x), independent across countries
- $n(s) \sim$ Geometric(ζ)

- Sector draws correlated across countries: $H_Z(z, z^*) = \mathcal{C}(F_Z(z), F_Z(z^*))$ with a **Gumbel copula**

$$\mathcal{C}(u, u^*) = \exp\left(-\left[(-\log u)^\rho + (-\log u^*)^\rho\right]^{\frac{1}{\rho}}\right), \quad \rho \geq 1$$

- Dependence measured by Kendall's $\tau(\rho) = 1 - 1/\rho$ (not to be confused with trade cost τ)
  - $\tau(\rho) = 0$: independent draws, dispersed import shares, low trade elasticity
  - $\tau(\rho) \rightarrow 1$: identical draws, little dispersion, high trade elasticity

Data and Concentration Facts

- Taiwan Annual Manufacturing Survey (2000, 2002–2004): universe of manufacturing establishments; sales by 7-digit product ($\approx$ 5-digit U.S. SIC); employment, materials, energy, revenue

Within-sector concentration, domestic producers

	median	mean
# producers/sector	10	25
inverse Herfindahl	3.9	7.3
share of top producer	0.40	0.45
sales, top to median	17	42

Size distribution of producers

fraction accounted for by	sales	wages
top 1%	0.41	0.24
top 5%	0.65	0.47

Estimating θ and Calibration

- Model: inverse markup linear in market share

$$\frac{1}{\mu_i} = \frac{\gamma - 1}{\gamma} - \left(\frac{1}{\theta} - \frac{1}{\gamma}\right)\omega_i$$

- Estimate μ_i with De Loecker and Warzynski (2012)
- Data slope ≈ -0.68; with $\gamma = 10$: $\frac{1}{\theta} = 0.68 + 0.10$, so $\theta \approx 1.28$

	Data	Model
median inverse HHI	3.9	3.8
median share top producer	0.40	0.41
median share	0.005	0.006
p75 share	0.02	0.03
p95 share	0.19	0.27
p99 share	0.59	0.59
inverse markup on share	-0.68	-0.68
aggregate import share	0.38	0.38
fraction exporters	0.25	0.25
trade elasticity	4	4

Parameters: $\gamma = 10$, $\theta = 1.28$, $\xi_x = 4.5$, $\xi_z = 0.6$, $\zeta = 0.04$, $f_d = 0.004$, $f_x = 0.21$, $\tau = 1.13$, $\tau(\rho) = 0.93$

Markup Distribution

	Data	Model	Autarky
aggregate markup	–	1.31	1.35
mean markup	1.13	1.14	1.15
median markup	1.11	1.12	1.12
p75 markup	1.12	1.14	1.14
p90 markup	1.15	1.21	1.23
p95 markup	1.20	1.31	1.35
p99 markup	1.48	1.67	1.76
std. dev. log markup	0.06	0.08	0.10
log p95/p50	0.08	0.16	0.19

- Aggregate (cost-weighted) markup well above the mean: large firms have high markups
- Moving from autarky to Taiwan's openness lowers the level and the dispersion of markups, mostly in the right tail

Gains from Reducing Trade Costs

Constant markups: gains only from first-best productivity. Variable markups: also reduced misallocation.
Pro-competitive gains $\equiv$ total gains $-$ first-best gains.

change in import share (%)	0 to 10	10 to 20	20 to 30	$\cdots$	0 to Taiwan
change in A, %	3.1	2.8	3.3		12.0
change in first-best A, %	1.9	2.5	3.1		10.2
pro-competitive gains, %	1.2	0.3	0.2		1.8
misallocation / autarky	0.86	0.83	0.80		0.79
change in markup, %	-1.6	-0.6	-0.4		-2.8
domestic	-1.4	-0.5	-0.5		-2.7
import	15.8	0.2	0.3		16.5
change in dispersion, %	-0.9	-1.1	-0.7		-2.8
domestic	-1.6	-1.6	-1.4		-5.8
import	23.4	0.3	-0.5		23.2
trade elasticity (ex post)	4.2	4.1	4.0		4.0

Reading the Table: Domestic vs. Import Markups

- Pro-competitive gains are about 1/6 of total gains (1.8 of 12.0), relatively more important near autarky; misallocation falls by about 20%
- The overall effect depends on markup responses in *both* domestic and import markets
  - Home firms lose domestic market share: lower markups and less dispersion
  - Foreign firms gain share in Home: higher markups and more dispersion
- Looking only at domestic markups is insufficient to determine the change in misallocation (the same lesson as ACDR's direct vs. indirect effects)

Importance of Head-to-Head Competition

- Matching a trade elasticity of 4 requires $\tau(\rho) = 0.93$
- High correlation $\Rightarrow$ weak comparative advantage $\Rightarrow$ genuine head-to-head competition after liberalization
- Lower correlation: strong comparative advantage, larger total gains but weak or even negative pro-competitive gains

$\tau(\rho)$	trade elast.	pro-comp., %	total, %
1.0	4.41	2.0	11.1
0.9	3.77	1.7	12.6
0.8	2.91	1.4	16.5
0.7	2.22	1.2	21.3
0.6	1.77	0.9	26.7
0.5	1.44	0.7	32.9
0.4	1.19	0.5	39.8
0.3	1.01	0.2	47.8
0.2	0.86	-0.1	57.4
0.1	0.74	-0.5	69.3
0.0	0.66	-0.9	86.1

Discussion and Takeaways

- Evidence on import-share dispersion suggests $\tau(\rho)$ between 0.8 and 1.0
- Correlation *per se* is not the issue. The real question: does liberalization raise competitive pressure on hitherto dominant firms, or open new opportunities to charge high markups abroad?
  - The latter is more likely with strong comparative advantage: a trade-off between standard and pro-competitive gains
  - Correlation is sufficient, not necessary, for weak comparative advantage; e.g. a country that is *small* relative to the world faces strong competition even with independent draws
- Across this lecture
  - Non-CES demand delivers variable markups, but with Pareto + gravity the pro-competitive gains are zero (homothetic) or negative (ACDR)
  - Pro-competitive gains require a change in the *joint* distribution of markups and market shares; foreign markups matter as much as domestic ones
  - Oligopoly with head-to-head competition (EMX) can generate sizable gains, especially starting near autarky

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